Grounding biological data to databases

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Abstract

Marker genes are used to identify cell types in single-cell studies, but they often fail to replicate across studies. One reason is how markers are reported: the common cell type–gene format omits the comparisons that made a marker informative, even though those comparisons determine how it can be reused. We introduce `.onto`, a file format that stores source-grounded marker evidence, and `mrkr`, a tool that uses LLMs to extract marker statements and normalize their biological labels before deterministic mapping to stable database identifiers. On a manually curated benchmark, `mrkr` achieved a mean cell type–gene extraction F1 of 0.64, about fourfold higher than LLM marker generation or differential-expression-based marker selection. When we applied `mrkr` to 979 bioRxiv and Human Cell Atlas-linked papers, stable identifiers connected 248 cross-paper marker-panel pairs missed by exact label matching. However, identifiers alone did not make marker panels reproducible. Among 37 cell type labels reported in at least ten papers, 51.8% retained a shared marker after combining marker panels from two papers, but only 1.7% did after ten. Our formalization demonstrates that marker validity depends on the cell groups used for comparison, which can change across papers, yet only 5.2% of marker statements named those groups. In a controlled analysis of one Cell Annotation Platform expression matrix, expanding the comparison groups reduced recovery of reported markers from 74% to 21%. Together, these results motivate retaining both stable identifiers and source evidence, including explicit comparisons, when connecting markers across papers.

1 Introduction

Cell types are often identified by marker genes. In single-cell RNA-seq, researchers partition cells by gene expression and report genes that distinguish one group from another [Macosko et al., 2015, Stuart et al., 2019, Wolf et al., 2018, Heumos et al., 2023, Pullin and McCarthy, 2024]. These genes connect transcriptomic measurements to cell type labels used in experiments, annotation tools, databases, and atlases. Yet marker genes often fail to transfer between datasets, and cell-type-specific molecular signals can fail to replicate across cohorts of the same tissue or disease [Crow et al., 2018, Krzak et al., 2026].

Marker reuse can fail for several reasons. Assay chemistry, sequencing depth, quantification, normalization, replicate structure, donor variation, disease state, and differential-expression methods can change the measured signal [Soneson and Robinson, 2018, Squair et al., 2021, Hoerbst et al., 2025]. Additionally, any one study observes only a finite set of cells, so it cannot test a marker against groups that were not sampled. These limitations concern both the measurement and the experiment. A separate limitation concerns reporting. Papers and databases often reduce discovered markers to a cell type–gene pair and omit the source evidence and comparison that produced it [Kiselev et al., 2018, Aran et al., 2019, Pliner et al., 2019, Zhang et al., 2019, Ianevski et al., 2022, Conde et al., 2022, Abdelaal et al., 2019, Hu et al., 2022]. Once this information is removed, later users cannot determine whether marker failure arose from measurement variation, a new cell group, or an unsupported comparison.

This reporting scheme makes marker gene transfer challenging. Marker genes are measured locally, and cell type labels are used to transfer them globally. Cross-study reuse requires that authors refer to the same genes, cell types, tissues, and organisms with stable identifiers. It also requires that each marker retain the cell groups against which it was supported. Cell ontologies and nomenclature systems provide the first link [Miller et al., 2020, Osumi-Sutherland et al., 2021, Tan et al., 2026], but they do not provide the second. A gene can distinguish a T cell group from B cells and fail to distinguish it from another T cell subtype even when every group has a stable identifier.

We developed `mrkr` and the `.onto` format to recover reported markers from papers and ground them in stable identifiers (**Figure 1**). An LLM extracts the reported evidence and converts author language into normalized biological labels. Deterministic lookup then assigns gene, cell type, tissue, and organism identifiers, and validation connects every retained statement to the exact source text. We evaluate this workflow against `LLMarkers`, a manually curated benchmark of 1,560 cell type–gene pairs

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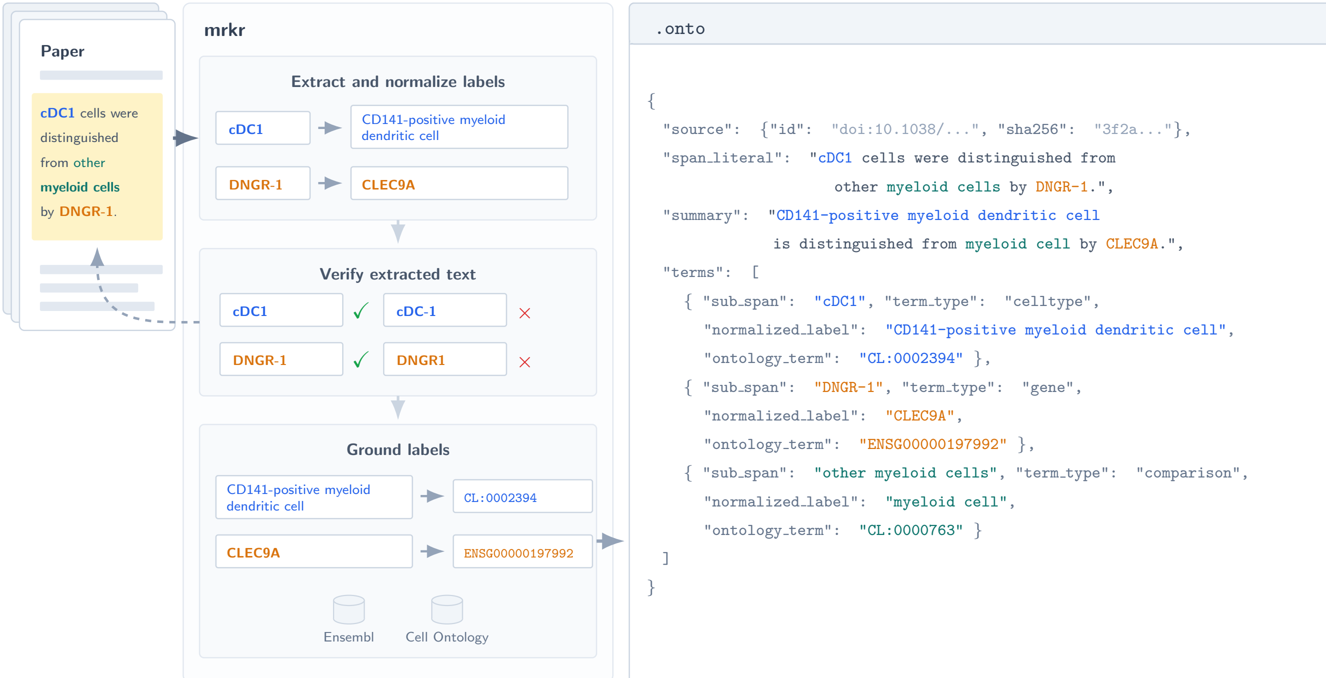

Figure 1: `mrkr` converts reported marker statements into machine-readable `.onto` records. The LLM extracts the source sentence and biological terms, then proposes normalized labels. Exact-string validation accepts `cDC1` and `DNGR-1` and rejects `cDC-1` and `DNGR1`, which do not occur in the source. Deterministic lookup maps normalized gene and cell type labels to Ensembl and Cell Ontology identifiers; the LLM does not assign identifiers. The `.onto` record retains the source sentence, normalized statement, source terms, and identifiers. Blue denotes the cell type, orange the marker gene, and teal the comparison group.

from seven single-cell RNA-seq studies.[2] We then apply it to 979 bioRxiv and Human Cell Atlas-linked papers. The resulting corpus lets us test which records stable identifiers can connect, how reported marker panels vary across papers, and which information remains missing. We use Cell Annotation Platform data and a formal model to show why comparison groups are necessary for interpreting those connections.

2 Results

`mrkr` maps labels stable identifiers

Reported marker genes are distributed across manuscript text, figures, and supplementary tables. `mrkr` converts manuscript text into a validated `.onto` file. Each selected marker statement links the exact source text to one organism, one cell type label, at least one marker gene, marker direction, and any tissue or comparison named in the source. The LLM returns normalized labels. Deterministic grounding then maps those labels to Ensembl, NCBI Taxonomy, Cell Ontology, and UBERON identifiers. Validation checks the manuscript hash, text offsets, term offsets, and required term counts.[3] Importantly, the LLM does not choose the final database identifier but rather it normalizes the label for lookup. This separation lets normalized author language be mapped, audited, and revised without changing the underlying evidence.

Markers are not recoverable from DEGs

To assess the importance of normalizing and recovering marker genes from papers and supplementary data, we built `LLMarkers`. The benchmark contains 2,528 manually curated source records from seven scRNA-seq studies and six tissues. These records represent 2,123 marker terms, 168 cell types, 641 genes, and 1,560 unique cell type–gene pairs (**Supplementary Table S1**).[4] Most evidence was not available in manuscript text alone. Of the 1,560 cell type–gene pairs, 1,151 (74%) appeared only in figures, 256 (16%) appeared only in text, and 153 (10%) appeared in both.[5]

Reported markers were not identical to the genes in the linked differential-expression (DE) tables. The fraction of curated markers found in these tables ranged from 18% in the study of human embryonic skeletal development [He et al., 2021] to 97% in the study of idiopathic pulmonary fibrosis [Adams et al., 2020]. In the Emont adipose study, the authors report *PDGFRA* as a marker of `hASPC1` through `hASPC6`, but the subpopulation-specific DE table included it only for `hASPC1`.[8] The reported strength and direction of a marker also depended on the set of cells compared. Among 544 curated cell type–gene pairs found in at least two differential-expression tables from the same study, 46 (8.5%) were upregulated in one table and downregulated in another.[9]

Marker gene DE rank was only partially consistent

A **LLMarkers benchmark**

Paper	CTs	Genes	Pairs	Tissue
Emont 2022	45	105	346	adipose
Hildreth 2021	31	141	234	adipose
He 2021	27	75	370	bone marrow
Gautam 2021	32	94	251	retina
Adams 2020	20	124	135	lung
Wagner 2020	7	62	58	ovary
Shamis 2020	19	125	171	testis
Total	**168**	**641**	**1,560**	**6 tissues**

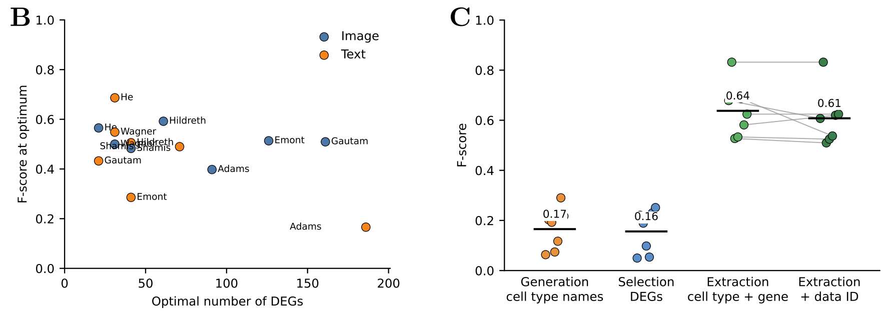

Figure 2: Reported markers are not fully recovered by DEG rank but can be extracted from source text. A. Cell types, marker genes, cell type–gene pairs, and tissues represented in the seven-study `LLMarkers` benchmark. **B**. Optimal DEG cutoff and corresponding F1 for recovering markers reported in text or images. **C**. F1 for generation from cell type names, selection from ranked DEGs, and joint extraction from manuscript text and candidate DEG sources. Extraction was scored as exact cell type–gene pairs and as exact `data_id`–cell type–gene triples. Each point represents one study; black bars show study means, and gray lines join extraction scores from the same study.

with reported markers in a paper. 53% of text markers and 50% of figure markers were among the top 50 differentially expressed genes. The top 100 DEGs recovered 66% and 62%, respectively, of reported markers.[10] The rank cutoff that maximized F-score also varied by study. Across studies, the optimal cutoff averaged 60 genes for text markers and 76 genes for figure markers, with mean peak F-scores of 0.44 and 0.51 (**Figure 2B**).[11] The Hildreth study shows that these rank and precision–recall distributions vary by cell type, with some DEGs being more consistent with reported marker sets than others (**Supplementary Figure S1**). Although differential expression is often described as a marker selection process, it does not fully reproduce what authors report as markers.

LLMs effectively extract markers

To test whether an LLM could be used to supplement marker selection from DEGs, we compared three marker gene curation tasks. We had the LLM generate markers from cell type names, select markers from ranked differential-expression tables, and extract marker statements and normalized labels from manuscript text while selecting the supporting differential-expression source in that paper (**Figure 2**; **Supplementary Table S2**). Generation and selection test the model's biological knowledge, whereas extraction tests whether the model can structure paper-specific evidence for later mapping.

LLMs recovered marker evidence more accurately than they generated or selected it. Generation from cell type names achieved a mean cell type–gene F1 of 0.17 against the human annotations.[12] Selection from ranked differential-expression tables achieved a mean F1 of 0.16 at the best input depth for each study.[13] By contrast, extraction used the paper rather than model memory. After source validation, gene mapping, and merging statements supported by the same source text, 152 marker statements containing 311 mapped marker terms remained. Five unsupported or ambiguous associations were excluded.[14] Mean exact cell type–gene F1 was 0.64. Requiring the correct supporting differential-expression source identifier produced a mean F1 of 0.61.[15] The useful LLM output is normalized, source-grounded text. Stable biological identifiers are assigned afterward by deterministic grounding. This result supports using LLMs to normalize and ground reported markers rather than replace biological expertise in selecting them. We implemented this workflow in `mrkr`.

Stable IDs connect markers across papers

Having established that `mrkr` can recover source-grounded marker evidence, we asked whether stable identifiers make that evidence reusable across papers. We applied `mrkr` to 504 bioRxiv preprints and 475 Human Cell Atlas-linked papers. All 979 `.onto` files passed source and structure validation. Of these papers, 721 contained at least one marker statement. The corpus contains 11,336 source-grounded marker statements and 27,713 marker-gene terms. Deterministic grounding assigned candidate identifiers to 26,919 marker-gene terms, 9,179 of the 11,336 cell type terms, and 450 of 603 extracted comparison terms. Only 586 marker statements (5.2%) include a comparison term.[17] Positive and negative markers were kept separate.

For each paper, we combined the mapped positive genes reported for one normalized cell type label. We call this set the reported marker gene panel. Requiring at least three genes per panel and excluding pairs drawn from the same article produced 3,454 reported marker panels and 10,437 pairs of panels from different articles that used the same normalized cell type label.[18]

We retained a Cell Ontology mapping only when the reported label matched the canonical label or an official exact synonym. This conservative rule retained 406 of

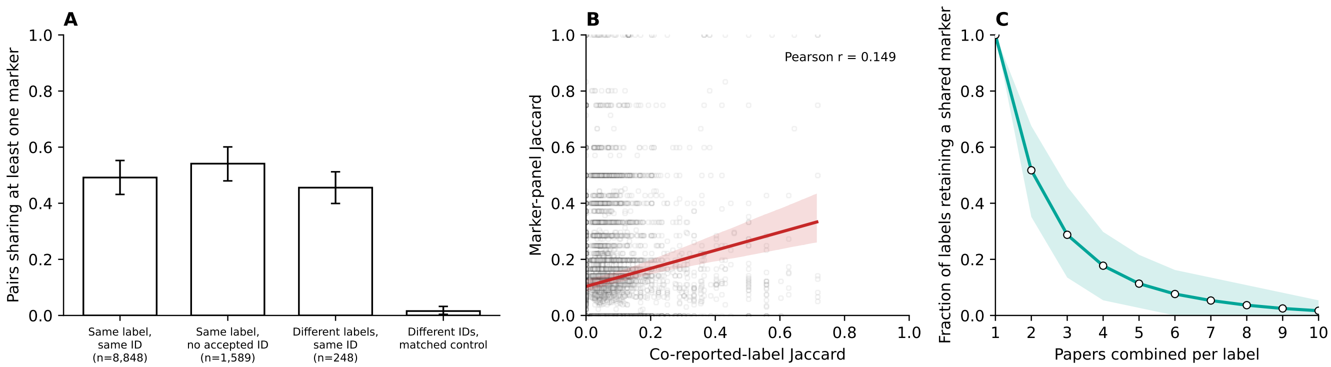

Figure 3: Stable identifiers connect marker evidence across papers, but reported marker panels remain variable. Each marker panel contains at least three mapped positive genes reported for one cell type in one paper. **A**. Fraction of cross-paper panel pairs sharing at least one marker for the same label and identifier, the same label without an accepted identifier, different labels with the same identifier, and a matched different-identifier control. Counts below the first three bars give the observed pairs. Identifier-only and control pairs were matched by corpus collection and panel size. Whiskers are 95% intervals from 1,000 bootstrap or matched-sampling replicates. **B**. Marker-panel Jaccard versus co-reported-label Jaccard for 10,437 cross-paper pairs with the same normalized target label. The target was excluded from the co-reported-label calculation. The red line is an ordinary least-squares fit; the band is a 95% interval from 1,000 bootstrap replicates that resampled target labels. **C**. Fraction of 37 normalized cell type labels retaining at least one shared marker as panels from additional papers are intersected. Points are means from 1,000 label-bootstrap and random-order replicates; the band spans the 2.5th–97.5th percentiles.

3,355 distinct reported-label mappings, representing 363 Cell Ontology identifiers. After requiring at least three markers per panel, 131 identifiers supported cross-paper comparisons. Twelve of these identifiers connected 248 marker-panel pairs that used different normalized labels. Examples include "plasma B cell" and "plasma cell," and "red blood cell" and "erythrocyte."[19]

Labels and identifiers provided overlapping but distinct links (**Figure 3**A). Exact labels connected 10,437 cross-paper panel pairs. Of these, 8,848 also shared an accepted identifier and 1,589 had no accepted identifier. Stable identifiers added 248 pairs that used different labels; 113 (45.6%) shared at least one marker. After matching by corpus collection and marker-panel size, marker sharing was 49.2% for same-label pairs with the same identifier, 54.1% for same-label pairs without an accepted identifier, and 1.6% for pairs with different identifiers.[20] Stable identifiers recover related marker evidence hidden by label variation, while reported labels retain evidence not yet linked to an accepted identifier. Both should be retained. Neither establishes that the sampled cell groups are biologically identical.

Other normalized cell type labels reported in each paper provide a limited proxy for the unreported comparison set of cells used to call a marker (**Figure 3**B). Across 10,437 pairs of papers that used the same normalized cell type label, marker-panel Jaccard was positively but modestly associated with co-reported-label Jaccard (Pearson $r = 0.149$).[21] Papers with more similar reported cell type contexts tended to report more similar marker panels, but co-reported labels do not identify the comparison used for a marker statement.

The shared marker intersection was rapidly lost as evidence accumulated (**Figure 3**C). We fixed the analysis to the 37 normalized cell type labels reported in at least ten papers and repeatedly added paper-level marker panels in random order. After combining two papers, an estimated 51.8% of labels retained at least one shared marker. This fraction fell to 28.8% after three papers, 11.3% after five, and 1.7% after ten. Repeating the analysis with 27 accepted Cell Ontology identifiers gave nearly identical results.[22] A cell type label or identifier therefore does not specify a marker gene panel that remains stable as papers accumulate.

Markers change with cellular context

Cross-paper differences can reflect changes in measurement, observed cell groups, or reporting. To isolate the effect of which cell groups are included, we tested the same marker genes against expression data from the HCA's Cell Annotation Platform (CAP). CAP contains a liver myeloid analysis nested within an all-cell liver analysis of the same cells. The local analysis separates ten related myeloid labels. The broader analysis adds non-myeloid cells and uses coarser author labels. This design holds the underlying myeloid cells constant while changing the groups included in the differential-expression calculation.

Reported markers were recovered more often in the partition from which they were curated (**Figure 4**). Across the ten local myeloid labels, local one-vs-rest differential expression recovered a mean of 74% of reported markers in the top 100 genes. The all-cell analysis

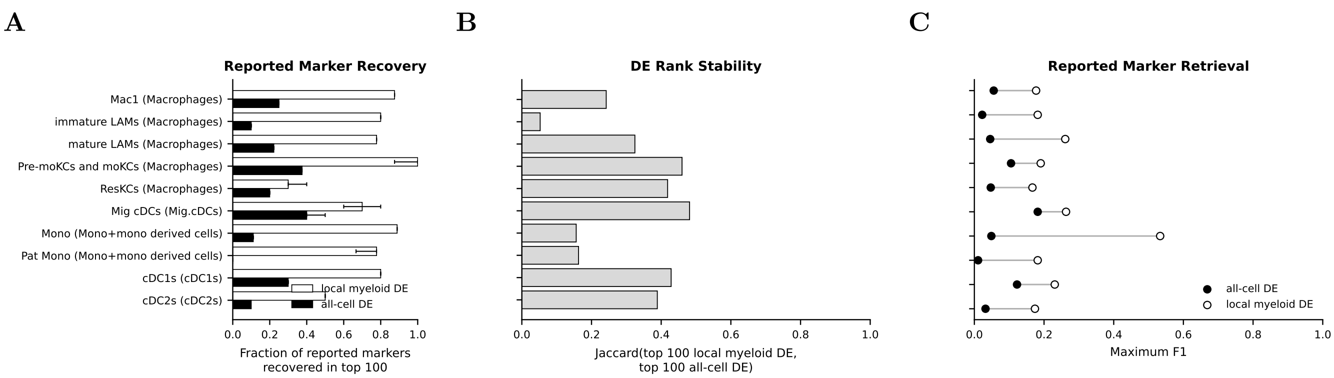

Figure 4: Marker recovery changes when the comparison groups change. The same liver myeloid cells were analyzed in a local myeloid partition and a broader all-cell liver partition. **A**. Fraction of reported markers recovered among the top 100 one-vs-rest DEGs. Whiskers are 95% intervals from 50 cell-type-stratified subsampling replicates. **B**. Jaccard similarity between the top 100 local and all-cell DEG rankings. **C**. Maximum F1 for recovering reported markers from each ranking. All panels show the ten labels from the local myeloid analysis.

recovered 21%. Recovery was higher in the local analysis for every label.[24] Despite the myeloid cells being the same, the marker ranking changed when additional context-relevant cells were added.

Stable IDs make markers comparable

The benchmark, corpus, and CAP results show that cell type labels and marker genes provide different information (**Supplementary Figure S2**). A marker gene panel must specify the target cell group and the groups it is expected to distinguish. Stable identifiers make those groups joinable across records. We formalized this statement mathematically. For a study s, partition p, cell group t, and comparison group h, let $M_{s,p}(t,h)$ be the genes called as markers for t against h. For a set of comparison groups C,

$$M_{s,p}(t;C) = \bigcap_{h \in C} M_{s,p}(t,h). \tag{1}$$

A single marker gene must pass every comparison in C. A marker gene panel P may use different genes for different comparisons, but it must contain at least one gene for each comparison. Thus, $P \cap M_{s,p}(t,h) \neq \varnothing$ for every $h \in C$.

The formalization proves that marker gene sets are intersections of pairwise marker sets, marker gene panels are hitting sets over the comparisons, and adding a comparison can only remove single-gene markers or add requirements to a panel.[25] A pooled one-vs-rest test and the intersection of all pairwise tests both use every non-target group, but they are not equivalent. The pooled test averages the background and can pass when one pairwise comparison fails. The formal result also explains hierarchical markers. A marker that passes a broad set of comparisons remains valid when the set is restricted, but a marker found among closely related cell groups need not identify the broader parent group [Domcke and Shendure, 2023, Beltz et al., 2026].

The formalization explains the role of stable identifiers. They make t and h consistent and joinable across papers, but they do not make a marker universal. When a marker statement names a comparison group, `mrkr` can normalize it and assign a stable identifier. Most statements in the corpus did not name that group, so retrospective curation cannot recover the missing comparison. A source-grounded `.onto` file provides the first required layer by preserving stable identifiers for the paper, cell type label, marker genes, tissue, organism, and any stated comparison together with the exact source text. We release the 979 validated `.onto` files, their validation metadata, and a derived SQLite database as a versioned corpus.[26] Future comparison-aware records can build on these identifiers without changing the source evidence.

3 Discussion

Marker reuse is limited in multiple ways: the assay or analysis may change the measured signal, a study may omit a relevant cell group, and a paper may omit the comparison that supported the marker. The first two can require new data. The third is avoidable, but a flat cell type–gene pair removes the information needed to assess these limits.

Our results separate the cell type label from the marker panel used to identify the reported cell type. Normalized cell type labels and stable Cell Ontology identifiers connect evidence that exact label matching misses, but a shared identifier does not imply a shared marker panel. Shared marker intersections rapidly disappeared as papers accumulated, including when labels were linked by accepted Cell Ontology identifiers. In CAP, changing the observed cell groups changed marker rankings within the same expression matrix. Importantly, stable identifiers establish which entities can be compared and the observed comparisons establish what the marker evidence supports.

LLMs can normalize variable author language for deterministic grounding in biological databases. They recovered source-grounded statements better than they generated or selected markers. This supports LLM-assisted language normalization rather than using an LLM as an independent authority on marker genes. The exact source text remains the evidence, and the database mapping remains auditable. Related language-model approaches inherit the same limit when the original comparison is absent [Hou and Ji, 2024, Levine et al., 2023, Zhao et al., 2024, Schaefer et al., 2025, Yang et al., 2025, Kedzierska et al., 2025].

We release the LLMarkers corpus of 979 papers. It connects reported markers to stable identifiers and the exact source text that supports each marker statement. However, it does not contain every marker that was tested or reported. Authors select which results to discuss, the large-scale extraction does not yet include figures, and only 5.2% of retained statements contain a comparison term. Marker recurrence therefore measures published support for reuse, not expression-based sensitivity or specificity.

Reporting cannot remove assay variation or supply cell groups that were never observed. It can prevent missing context from being confused with those limitations. Source-grounded target and comparison identifiers, combined with pairwise marker evidence or sufficient statistics, make comparison-defined marker sets reconstructable and missing comparisons explicit. This is what reporting can fix.

A Supplementary Material

Table S1: LLMarkers benchmark. The number of unique cell types (CTs), genes (Ensembl IDs), and (cell type, gene) pairs Overall, and across Text, Image, and DEG tables. The %DEG reports the percentage of pairs found in the study's DEG lists.

	Overall			Text				Image				DEG
Dataset	CTs	Genes	Pairs	CTs	Genes	Pairs	DEG%	CTs	Genes	Pairs	DEG%	Pairs
`Adipose (Emont)`	45	105	346	20	23	30	80	44	104	344	39	43,471
`Adipose (Hildreth)`	31	141	234	21	53	66	92	31	124	207	96	46,357
`Bone (He)`	27	75	370	26	69	98	53	24	29	309	13	5,382
`Eye (Gautam)`	32	94	251	12	22	24	88	28	82	235	49	24,998
`Lung (Adams)`	20	124	135	6	33	33	97	20	99	109	97	34,760
`Ovary (Wagner)`	7	62	58	7	46	37	89	6	54	41	80	14,357
`Testis (Shamis)`	19	125	171	15	91	123	78	11	59	64	94	9,663
Total	168	641	1,560	100	305	409	78	151	495	1,304	53	169,687

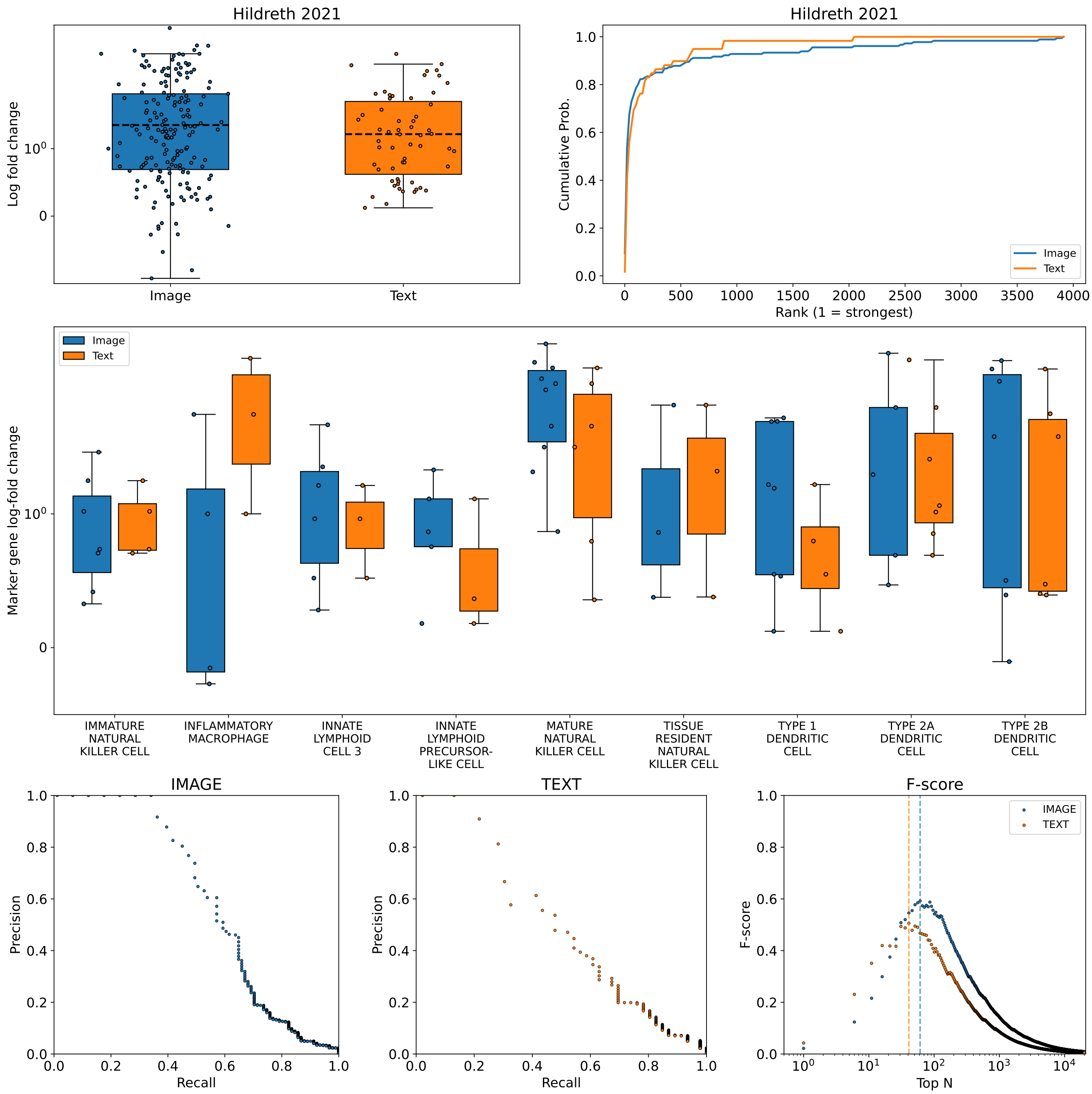

Figure S1: Reported marker strength and recovery vary by cell type in Hildreth et al. [2021]. Top: log-fold-change distributions and cumulative DEG-rank distributions for markers reported in images (blue) and text (orange). Middle: log-fold-change distributions stratified by cell type and source. Bottom: precision–recall curves and F1 across top-N DEG thresholds.

Table S2: LLM marker gene curation. All runs used Claude (claude-opus-4-6). Precision (P), recall (R), and F1 were computed against positive human text annotations. N: unique mapped pairs produced. %DEG: produced triples found in the matching DEG table. Rank: median DEG rank of matched triples. Opt N: rank threshold maximizing F-score. **(a)** Generation from cell type names. Pred. Acc.: cell types identified from anonymized gene lists. **(b)** Selection from ranked DEGs. Best N: input depth maximizing pair F1. **(c)** Joint extraction from manuscript text and the catalog of candidate DEG sources. Pair scores require the exact normalized cell type and gene; triple scores also require the exact supporting `data_id`. Retained claims passed current `mrkr` exact-source validation. Bottom rows show means across datasets.

(a) Generation

Dataset	Pred. %	N	%DEG	Rank	Opt N	P	R	F1	Cost ($)	Time (min)
`Adipose (Emont)`	13	317	–	–	–	0.03	0.37	0.06	0.48	3.2
`Adipose (Hildreth)`	10	249	–	–	–	0.13	0.48	0.20	0.41	2.8
`Bone (He)`	4	203	–	–	–	0.14	0.30	0.19	0.34	2.3
`Eye (Gautam)`	13	243	–	–	–	0.04	0.37	0.07	0.39	2.6
`Lung (Adams)`	40	165	–	–	–	0.07	0.48	0.12	0.29	2.0
`Ovary (Wagner)`	29	69	–	–	–	0.26	0.33	0.29	0.12	0.8
`Testis (Shamis)`	16	144	–	–	–	0.20	0.24	0.22	0.25	1.7
Mean	18	199	–	–	–	0.13	0.37	0.17	0.33	2.2

(b) Selection

Dataset	Best N	N	%DEG	Rank	Opt N	P	R	F1	Cost ($)	Time (min)
`Adipose (Emont)`	200	371	97	17	106	0.03	0.33	0.05	1.53	4.4
`Adipose (Hildreth)`	400	333	98	20	106	0.13	0.67	0.22	0.50	4.2
`Bone (He)`	300	293	97	26	116	0.13	0.38	0.19	0.95	3.4
`Eye (Gautam)`	20	218	95	8	21	0.06	0.44	0.10	0.43	2.7
`Lung (Adams)`	200	349	97	53	196	0.03	0.43	0.05	1.24	3.9
`Ovary (Wagner)`	200	110	100	28	76	0.17	0.35	0.23	0.35	1.3
`Testis (Shamis)`	300	227	99	32	181	0.19	0.36	0.25	0.93	2.8
Mean		272	98	26	115	0.11	0.42	0.16	0.85	3.2

(c) Extraction

Dataset	Sources	Claims	Gene terms	Cell type–gene			+ data ID		
				P	R	F1	P	R	F1
`Adipose (Emont)`	6	19	26	0.73	0.63	0.68	0.65	0.57	0.61
`Adipose (Hildreth)`	4	17	47	1.00	0.71	0.83	1.00	0.71	0.83
`Bone (He)`	5	34	57	0.78	0.40	0.53	0.73	0.39	0.51
`Eye (Gautam)`	7	24	33	0.48	0.59	0.53	0.48	0.57	0.52
`Lung (Adams)`	8	12	44	0.52	1.00	0.69	0.41	0.78	0.54
`Ovary (Wagner)`	3	10	35	0.81	0.45	0.58	0.65	0.59	0.62
`Testis (Shamis)`	6	36	69	0.89	0.48	0.62	0.89	0.48	0.62
Mean		22	44	0.75	0.61	0.64	0.69	0.59	0.61

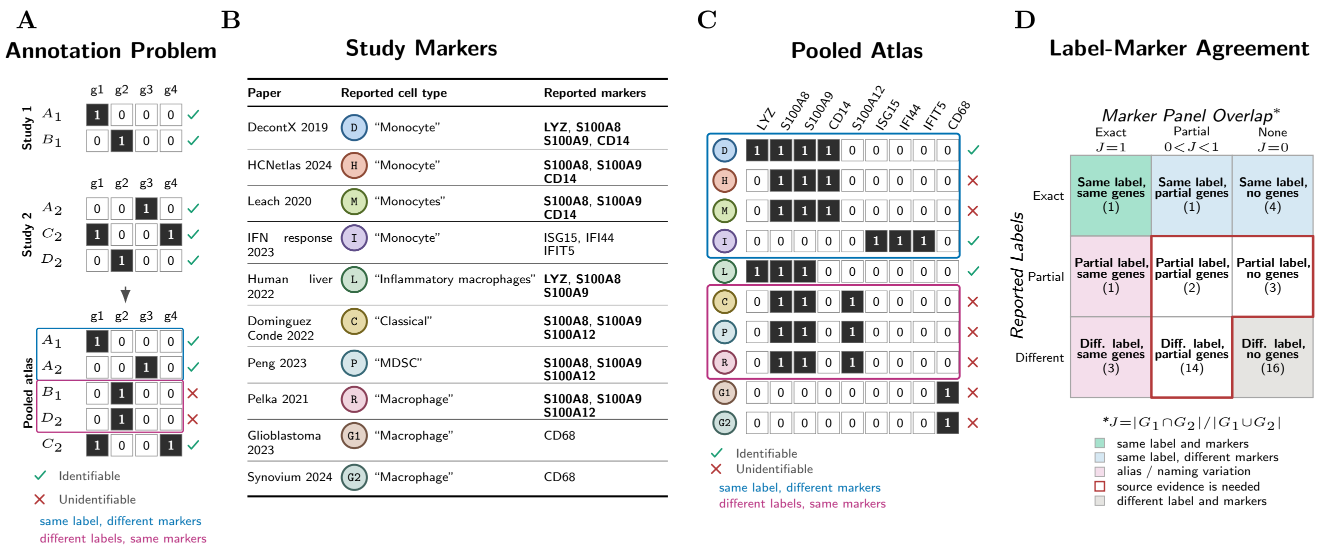

Figure S2: Marker genes and cell type labels provide different information. **A**. Binary marker matrices illustrate how a marker selected within one study can fail after studies are combined. Colored outlines indicate identical labels with different markers and different labels with identical markers. **B**. Source-grounded myeloid examples; bold genes are shared by at least two reported cell groups. **C**. Binary marker matrix for the examples in **B**. **D**. Relationship between label similarity and marker-panel similarity among the displayed examples. Marker overlap is Jaccard similarity, $J = |G_1 \cap G_2|/|G_1 \cup G_2|$.

B Methods

B.1 Benchmark construction and verification

The LLMarkers benchmark was assembled by manual curation of seven single-cell RNA-seq studies spanning six tissues: `adipose` [Emont et al., 2022, Hildreth et al., 2021], `bone` [He et al., 2021], `eye` [Gautam et al., 2021], `lung` [Adams et al., 2020], `ovary` [Wagner et al., 2020], and `testis` [Shami et al., 2020]. The benchmark was designed as a carefully linked evaluation set rather than a comprehensive survey of single-cell reporting styles. For each study, a curator read the manuscript and recorded each marker gene–cell type association, its verbatim sentence or figure evidence, source type, linked differential-expression table when available, organism, and Ensembl gene identifier. We preserved the original flat records and converted them losslessly into separate text statements, image statements, and differential-expression tables. Repeated source rows remain traceable through record identifiers rather than being discarded. Ten mouse records from the Adams study remain in the all-organism benchmark totals; the current extraction evaluation is restricted to human records.[31]

Each study's supplementary tables were parsed to extract differential expression gene lists. Header detection was performed automatically by scanning the first 10 rows for a row containing expected column names (e.g., gene, p_value, avg_log2FC). The resulting DEG lists were stored in the same evidence schema as human annotations, enabling direct comparison.[32]

We verified the converted benchmark against both the source records and manuscript text. Every source row round-tripped exactly and every source-file hash was unchanged. Of 166 primary text evidence claims, 158 (95%) were anchored to exact manuscript offsets; the remaining eight are retained with an explicit unanchored status rather than treated as verified text.[33]

B.2 Marker strength analysis

To quantify where human-curated marker genes fall in the differential expression ranking, we matched each marker to its corresponding entry in the dataset's DEG list. For human-curated markers, matching was performed on the four-part key (dataset, data_id, group_name, feature_id), requiring exact agreement on the cell type, gene, and supplementary data source. For LLM-generated markers, which have no data_id, we matched on (group_name, feature_id) against all DEG data sources within each dataset. Markers that could not be matched—because the gene was absent from the DEG list or lacked an Ensembl ID—were excluded from this analysis.[34]

For each matched marker, we recorded its rank within the cell type's DEG list (rank 1 = highest log-fold change) and its log-fold change. We computed the fraction of markers falling within the top N DEGs for thresholds $N \in \{10, 25, 50, 100\}$ and separately tabulated these fractions for text-sourced markers, image-sourced markers, and generated markers.[35]

To test whether discussed markers are drawn from stronger DEGs than expected by chance, we performed permutation tests at the level of individual cell types within each dataset and data source. For each cell type with at least two discussed markers, we sampled a size-matched set of genes from the top 100 significant ($p < 0.05$) DEGs that were not in the human-curated marker set. We then compared the mean rank of the discussed markers against the null distribution obtained by randomly partitioning the combined set 10,000 times. Tests were two-sided at $\alpha = 0.05$.[36]

To estimate the optimal DEG list depth for recovering each marker set, we computed precision–recall curves by varying a rank threshold N from 1 to 20,000. At each threshold, we defined the top N DEGs as predicted markers and computed precision, recall, and the F-score. The optimal N was defined as the threshold that maximized the F-score. This analysis was performed separately for each dataset and marker source (text, image, or generated).[37]

Differences in rank and log-fold change between marker sources were assessed using two-sided Mann–Whitney U tests.[38]

B.3 LLM generation and extraction

We evaluated three approaches to LLM-based marker gene curation across the seven benchmark datasets. All experiments used Claude (claude-opus-4-6) with temperature 0 and a maximum output length of 32,000 tokens; we did not benchmark model-to-model variation. Costs were computed from the model's API pricing ($5 per million input tokens, $25 per million output tokens for claude-opus-4-6).[39]

In the generation experiment (celltypes-to-genes), we gave the model the list of cell type names from each dataset's human-curated marker set and asked it to generate 5–15 well-established marker genes per cell type. The model received only the species and cell type names—no manuscript text, figures, or supplementary data. In a second generation experiment (genes-to-celltypes), we extracted the gene list for each cell type from the human-curated set, replaced the cell type names with anonymous labels (Group 1, Group 2, . . .), and asked the model to predict the cell type for each group. We measured prediction accuracy as the fraction of exact string matches between predicted and original cell type names. Non-exact predictions were counted as related when they showed substring containment or shared label tokens; remaining predictions were counted as having no lexical relation.[40]

In the selection experiment, we parsed each dataset's DEG tables, ranked genes per cell type by absolute log fold change, and presented the top N genes (with log fold change and adjusted p-value) to the model, asking it to select which are true markers. We swept $N \in \{10, 20, 30, 40, 50, 100, 200, 300, 400, 500\}$ for each dataset and reported the N that maximized pair-level F1. As a control for whether cell type identity aids selection, we repeated the full sweep with cell type names replaced by anonymous cluster labels (Cluster 1, Cluster 2, ...), preserving the gene lists and statistics but removing biological context from the names.[41]

To quantify the best possible selection performance, we computed a DEG-constrained max F1 score at each N. Let H be the set of human-curated marker pairs and D_N be the set of marker pairs available in the top-N DEG lists shown to the model. The recoverable set is $R_N = H \cap D_N$. Any selector restricted to D_N cannot recover pairs outside R_N. The maximum achievable pair-level F1 is therefore

$$\mathrm{F1}_{\max}(N) = \frac{2|R_N|}{|H| + |R_N|}, \tag{2}$$

which is attained by selecting exactly the recoverable set R_N (precision = 1). In **Figure 2**, the bound markers for selection are plotted at each dataset's selected operating point using that same dataset-specific N.

In the joint extraction experiment, Claude received the full manuscript and a catalog mapping each candidate `data_id` to the cell type labels present in that paper's DEG sources. For every reported human marker association, the model returned the cell type, gene, verbatim supporting text, and one candidate `data_id`, or null when no source was supported. The model did not receive the human annotations. When it returned null, an exact target-label match supplied the narrowest candidate source; this affected five of 152 retained claims. We converted the output to the current `mrkr` claim structure and coalesced records with the same source text and target into marker panels.[42]

The extraction evaluation used positive human text annotations. Validation required every retained claim to have an exact manuscript offset and every retained marker gene to occur explicitly in that source text. Unsupported genes and claims with ambiguous source offsets were excluded. Gene labels were mapped programmatically with the packaged `gmap.txt`. We computed macro-averaged precision, recall, and F1 across studies for exact normalized cell type–gene pairs and exact `data_id`–cell type–gene triples. The same extracted terms were used for both scores.[43]

The joint experiment saved each raw model response before validation. Every derived artifact records hashes for the prompt, manuscript, human benchmark, DEG catalog, and gene map. The run is therefore resumable and fails if an input changes.[44]

B.4 Paper corpora

We selected bioRxiv preprints from a local mirror of 428,741 MECA archives downloaded from the bioRxiv source repository in January 2025. Each archive contains JATS XML and structured metadata.[45] We required a CC-BY or CC0 license, terms that indicated human relevance, and terms related to cell types, single-cell RNA sequencing, or marker genes. We kept the most recent version for each DOI. These filters produced 504 bioRxiv manuscripts.[46] JATS XML was converted to Markdown while preserving section structure.

We assembled the second corpus from publications linked to Human Cell Atlas projects. Project metadata supplied publication titles, URLs, and DOIs. We deduplicated records by DOI and then by normalized title. Of 515 linked publication records, 475 had manuscript text that could be normalized to Markdown and processed.[47]

B.5 Label normalization and identifier grounding

The corpus source manifest lists one paper identifier, corpus collection, organism, manuscript path, and source hash per paper. We processed all 979 manuscripts with `mrkr` version 0.4.0 at the pinned source revision in `analysis/mrkr.lock`. Extraction used Claude (claude-opus-4-6). The model received the complete Markdown manuscript and the organism but could not assign database identifiers.

For each paper, `mrkr extract` returned exact source text, normalized labels, and term offsets. `mrkr ground` mapped human genes with the packaged gene map, organisms to NCBI Taxonomy, cell types and comparisons to Cell Ontology, and tissues to UBERON. `mrkr validate` then checked the source hash, source offsets, explicit term offsets, required term counts, gene direction, and grounding metadata.[48] Canonical Ensembl GRCh38 release 114 symbols take priority when a gene alias is ambiguous. Each saved `.onto` file records the model and response identifier, prompt digest, gene-map digest, ontology queries, selected identifiers, and validation state. The archive stores the raw model responses separately.

Some initial outputs contained individual statements or gene terms that failed exact-source validation. We repaired these files without another model call. The repair removed only the failing statement or conflicting gene term, recorded every removal, and reran grounding and validation.[49] Papers with no retained marker statement remained in the corpus as valid empty results. We built the analysis database from the final ontology manifest. The database stores paper metadata, marker statements, exact source text and offsets, normalized terms, identifiers, directions, and grounding state. Ontology hierarchy expansion was disabled for the primary analysis.

B.6 Cross-paper links by labels and identifiers

We analyzed mapped positive marker genes separately from mapped negative marker genes. A reported marker gene panel is the set of mapped positive genes assigned to one normalized cell type label in one paper. Labels were normalized with Unicode NFKC, case folding, replacement of non-word characters with spaces, and whitespace normalization. The primary analysis used exact normalized-label equality. It did not use substring, token-overlap, or semantic label matching. We retained panels with at least three mapped genes.

We grouped source records by normalized article title and excluded comparisons within the same article. For every cell type label reported in at least two articles, we compared all retained paper pairs. Marker similarity was the Jaccard similarity between the two sets of Ensembl gene identifiers. We also recorded whether the sets shared any gene and the fraction of each set recovered by the other.

For each pair of papers that used the same cell type label, we sampled one pair with different labels. Control panels matched the original corpus collection and the base-2 marker-count bin. We required different article keys and repeated the control sampling with 25 fixed random seeds.[50] Pair-weighted estimates give each paper pair equal weight. Label-balanced estimates first average the paper pairs for each label and then give each recurring label equal weight. We used 5,000 label-bootstrap replicates for the 95% intervals among papers that used the same label. Control intervals are the 2.5th and 97.5th percentiles across the 25 matched draws.

To measure cumulative marker-panel intersection, we fixed the cohort to normalized labels reported in at least ten papers. In each of 1,000 bootstrap replicates, we resampled labels with replacement, randomly ordered the paper-level panels for each selected label, and intersected the first one through ten panels. At each step, we recorded whether the intersection contained at least one marker. The plotted estimate is the mean fraction of labels with a nonempty intersection. Intervals are the 2.5th and 97.5th percentiles across replicates. We repeated the calculation for accepted Cell Ontology identifiers and obtained similar results. Separate sensitivity analyses repeated the pairwise and all-paper intersection summaries with minimum panel sizes of one, three, five, and ten genes.

To measure co-reported labels, we removed the shared cell type label from each paper's label set and computed Jaccard similarity between the remaining sets. We then compared this value with marker-panel Jaccard similarity. The fitted line is an ordinary least-squares fit. Its band is a 95% interval from 1,000 bootstrap replicates that resampled normalized target labels as clusters. Pearson correlation summarizes the unbinned association. Co-reported labels describe the labels in a paper. They do not reconstruct the cell-level comparison.

We performed a separate ontology analysis with Cell Ontology release `v2026-06-08`. A reported label was eligible only when it equaled the canonical label or an official exact synonym for its mapped term. We excluded broad, narrow, related, and matches supported only by full-text coverage. Identifier-recovered pairs shared an accepted Cell Ontology identifier but used different normalized labels. For every such pair, we sampled three references matched by corpus collection and the unordered pair of base-2 marker-count bins: a same-label pair with the same accepted identifier, a same-label pair without an accepted identifier, and a pair with different identifiers. Because grounding is deterministic from the normalized label, no same-label pair had two different accepted identifiers. We gave each identifier-recovered pair equal weight. Intervals are the 2.5th and 97.5th percentiles from 1,000 pair-bootstrap and matched-sampling replicates. An ontology-term-balanced analysis is retained as a sensitivity analysis.

The matrices in **Supplementary Figure S2** are schematic. The myeloid examples are source-grounded marker statements selected to illustrate relations between reported labels and marker gene panels. No corpus estimate is computed from these examples.[51]

B.7 Cell Annotation Platform expression analysis

We used CAP dataset 1440 as the local liver myeloid analysis and CAP dataset 1437 as the broader all-cell liver analysis. The files are nested, not independent. All 40,821 cell barcodes in dataset 1440 occur in the 167,598-cell all-cell file.[52] We loaded each file with `anndata.read_h5ad`, used the `user_provided` expression layer when present, and used `author_cell_type` as the label. We mapped each local myeloid label to its broader author label in the all-cell analysis.

For each label, we ranked all genes by the difference in mean expression between the cell group of interest and the remaining cells, divided by its standard error. Reported markers came from the CAP marker JSON and were restricted to genes mapped to human Ensembl identifiers. Top-100 recovery is the fraction of reported markers among the first 100 ranked genes. Differential-expression stability is the Jaccard similarity between the local and all-cell top-100 sets. Marker retrieval F1 treats the reported markers as positives and reports the maximum F1 over rank cutoffs.

For uncertainty in top-100 recovery, we performed 50 cell-type-stratified subsampling replicates. Each replicate sampled 80% of cells within each author label, with at most 10,000 cells per label. Figure 4A shows the 2.5th and 97.5th percentiles.

B.8 Lean formalization

We encoded the formal results in Lean 4 under `analysis/formal`. The binary model records studies, partitions, cell groups, pairwise comparisons, marker genes, and marker gene panels. The score model also records real-valued marker scores, thresholds, weighted backgrounds, and sufficient statistics.[53] We used `span` to align the Supplementary Note with Lean declarations and to inspect the dependency graph [Pachter, 2026].

The Lean development proves the marker-family intersection theorem, monotonicity under restriction of the comparison groups, local-to-global and hierarchy results, equivalence between the all-pair conjunction and the binary marker-family definition, marker panels as hitting sets, panel monotonicity, weighted-background identities, score-thresholded results, and examples in which a cell type–gene projection loses comparison information.[54] The Supplementary Note separates the intersection of pairwise tests from a pooled one-vs-rest statistic.

B.9 Data and code availability

Analysis code, the human benchmark, validated corpus archive, derived result tables, figure code, and Lean formalization are available at `https://github.com/sbooeshaghi/llmarkers`. The `mrkr` tool is available at `https://github.com/sbooeshaghi/mrkr`. The corpus archive contains the 979 validated `.onto` files, source and ontology manifests, raw model responses, repair reports, validation metadata, and the derived SQLite database. The original manuscripts are not redistributed when their licenses do not allow it. Each source manifest records the paper identifier and expected manuscript hash so that licensed source text can be restored and validated.

C Acknowledgements

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D Author contributions

A.S.B. conceived of the study, and compiled the first draft of the benchmark. N.A., along with A.S.B., completed the benchmark and performed initial analyses for verifying correctness of the `LLMarkers`. A.W. suggested expanding the analysis to the bioRxiv and Human Cell Atlas-linked corpora. A.S.B. performed the remaining analysis, along with Claude Opus 4.6, and wrote the manuscript. All authors met and discussed results and reviewed the manuscript.

E Disclosures

All em dashes were placed, in the text, by the author's own hands. Generative language models were used by the authors to check for grammatical errors in the text, fix bugs in code, and debug LaTeX issues.

Claim evidence registry

#	ID	Claim	Sources
1	`fig-mrkr-overview`	*Figure/table source record*	`docs/paper/biorxiv/tex-figures/fig1_mrkr_overview_body.tex` `README.md`
2	`intro-benchmark-summary`	`LLMarkers`, a manually curated benchmark of 1,560 cell type–gene pairs from seven single-cell RNA-seq studies.	`analysis/benchmark_verification.ipynb`
3	`mrkr-record-contract`	Each selected marker statement links the exact source text to one organism, one cell type label, at least one marker gene, marker direction, and any tissue or comparison named in the source. The LLM returns normalized labels. Deterministic grounding then maps those labels to Ensembl, NCBI Taxonomy, Cell Ontology, and UBERON identifiers. Validation checks the manuscript hash, text offsets, term offsets, and required term counts.	`analysis/mine_corpus.py` `analysis/build_claim_db.py` `README.md`
4	`benchmark-summary`	The benchmark contains 2,528 manually curated source records from seven scRNA-seq studies and six tissues. These records represent 2,123 marker terms, 168 cell types, 641 genes, and 1,560 unique cell type–gene pairs (**Supplementary Table S1**).	`analysis/artifacts/benchmark_evidence_v1/reconciliation.json` `analysis/benchmark_verification.ipynb`
5	`benchmark-modality`	Of the 1,560 cell type–gene pairs, 1,151 (74%) appeared only in figures, 256 (16%) appeared only in text, and 153 (10%) appeared in both.	`analysis/benchmark_verification.ipynb`
6	`fig-marker-selection`	*Figure/table source record*	`analysis/build_fig2_marker_recovery.py`
7	`fig-llm-curation`	*Figure/table source record*	`analysis/build_fig2_marker_recovery.py` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv` `analysis/evaluate_mrkr_joint_deg_extraction.py`
8	`benchmark-deg-overlap`	The fraction of curated markers found in these tables ranged from 18% in the study of human embryonic skeletal development [He et al., 2021] to 97% in the study of idiopathic pulmonary fibrosis [Adams et al., 2020]. In the Emont adipose study, the authors report *PDGFRA* as a marker of `hASPC1` through `hASPC6`, but the subpopulation-specific DE table included it only for `hASPC1`.	`analysis/benchmark_verification.ipynb`
9	`benchmark-sign-contradictions`	Among 544 curated cell type–gene pairs found in at least two differential-expression tables from the same study, 46 (8.5%) were upregulated in one table and downregulated in another.	`analysis/benchmark_verification.ipynb`
10	`expert-ranks`	53% of text markers and 50% of figure markers were among the top 50 differentially expressed genes. The top 100 DEGs recovered 66% and 62%, respectively, of reported markers.	`analysis/lfc_comparison.ipynb`
11	`expert-optimal-n`	Across studies, the optimal cutoff averaged 60 genes for text markers and 76 genes for figure markers, with mean peak F-scores of 0.44 and 0.51 (**Figure 2B**).	`analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
12	`gen-f1`	Generation from cell type names achieved a mean cell type–gene F1 of 0.17 against the human annotations.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
13	`sel-f1`	Selection from ranked differential-expression tables achieved a mean F1 of 0.16 at the best input depth for each study.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`

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#	ID	Claim	Sources
14	`ext-results`	152 marker statements containing 311 mapped marker terms remained. Five unsupported or ambiguous associations were excluded.	`analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv` `analysis/evaluate_mrkr_joint_deg_extraction.py`
15	`ext-text-f1`	Mean exact cell type–gene F1 was 0.64. Requiring the correct supporting differential-expression source identifier produced a mean F1 of 0.61.	`analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_summary.json` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv`
16	`fig-corpus-marker-reuse`	*Figure/table source record*	`analysis/analyze_mrkr_corpus.py` `analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_pairs.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_matched_summary.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/coreported_context_metrics.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/coreported_context_fit.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/label_intersection_accumulation.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/ontology_intersection_accumulation.tsv`
17	`corpus-source-grounded-summary`	All 979 `.onto` files passed source and structure validation. Of these papers, 721 contained at least one marker statement. The corpus contains 11,336 source-grounded marker statements and 27,713 marker-gene terms. Deterministic grounding assigned candidate identifiers to 26,919 marker-gene terms, 9,179 of the 11,336 cell type terms, and 450 of 603 extracted comparison terms. Only 586 marker statements (5.2%) include a comparison term.	`analysis/artifacts/mrkr_corpus_analysis_v1/corpus_summary.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/term_coverage.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/report.md`
18	`corpus-analysis-ready`	10,437 pairs of panels from different articles that used the same normalized cell type label.	`analysis/artifacts/mrkr_corpus_analysis_v1/label_profiles_analysis.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv`
19	`corpus-identifier-linkage`	This conservative rule retained 406 of 3,355 distinct reported-label mappings, representing 363 Cell Ontology identifiers. After requiring at least three markers per panel, 131 identifiers supported cross-paper comparisons. Twelve of these identifiers connected 248 marker-panel pairs that used different normalized labels. Examples include "plasma B cell" and "plasma cell," and "red blood cell" and "erythrocyte."	`analysis/artifacts/mrkr_corpus_analysis_v1/cell_ontology_label_audit.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/ontology_profiles_analysis.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_pairs.tsv`

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#	ID	Claim	Sources
20	`corpus-label-signal`	Exact labels connected 10,437 cross-paper panel pairs. Of these, 8,848 also shared an accepted identifier and 1,589 had no accepted identifier. Stable identifiers added 248 pairs that used different labels; 113 (45.6%) shared at least one marker. After matching by corpus collection and marker-panel size, marker sharing was 49.2% for same-label pairs with the same identifier, 54.1% for same-label pairs without an accepted identifier, and 1.6% for pairs with different identifiers.	`analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_pairs.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_matched_summary.tsv`
21	`corpus-coreported-context`	Across 10,437 pairs of papers that used the same normalized cell type label, marker-panel Jaccard was positively but modestly associated with co-reported-label Jaccard (Pearson $r = 0.149$).	`analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/coreported_context_metrics.tsv`
22	`corpus-marker-accumulation`	After combining two papers, an estimated 51.8% of labels retained at least one shared marker. This fraction fell to 28.8% after three papers, 11.3% after five, and 1.7% after ten. Repeating the analysis with 27 accepted Cell Ontology identifiers gave nearly identical results.	`analysis/artifacts/mrkr_corpus_analysis_v1/label_intersection_accumulation.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/ontology_intersection_accumulation.tsv`
23	`fig-cap-partition`	*Figure/table source record*	`analysis/build_cap_liver_local_global_deg.py` `analysis/artifacts/cap_liver_local_global_deg_summary.tsv` `analysis/artifacts/cap_liver_local_global_recovery_subsampling_summary.tsv`
24	`cap-liver-local-global-de`	Across the ten local myeloid labels, local one-vs-rest differential expression recovered a mean of 74% of reported markers in the top 100 genes. The all-cell analysis recovered 21%. Recovery was higher in the local analysis for every label.	`analysis/artifacts/cap_liver_local_global_deg_summary.tsv` `analysis/artifacts/cap_liver_local_global_recovery_subsampling_summary.tsv` `analysis/build_cap_liver_local_global_deg.py`
25	`formal-marker-definition`	The formalization proves that marker gene sets are intersections of pairwise marker sets, marker gene panels are hitting sets over the comparisons, and adding a comparison can only remove single-gene markers or add requirements to a panel.	`analysis/formal/MarkerIdentifiability/Basic.lean` `analysis/formal/CLAIMS.md`
26	`corpus-resource-release`	We release the 979 validated `.onto` files, their validation metadata, and a derived SQLite database as a versioned corpus.	`analysis/artifacts/mrkr_corpus_onto_v2.tar.gz.sha256` `README.md`
27	`tab-benchmark`	*Figure/table source record*	`analysis/benchmark_verification.ipynb`
28	`fig-hildreth-detail`	*Figure/table source record*	`analysis/lfc_comparison.ipynb`

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#	ID	Claim	Sources
29	`tab-llm-curation`	*Figure/table source record*	`analysis/ext_vs_gen.ipynb` `analysis/selection_analysis.ipynb` `analysis/llm_extraction_eval.ipynb` `analysis/artifacts/fig2_llm_recovery.tsv` `analysis/artifacts/fig2_llm_recovery_current.tsv` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_pair_review.tsv` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_triple_review.tsv` `analysis/build_fig2_marker_recovery.py` `analysis/evaluate_mrkr_joint_deg_extraction.py`
30	`fig-paper-celltype-joint`	*Figure/table source record*	`docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint_body.tex`
31	`methods-benchmark-schema`	The LLMarkers benchmark was assembled by manual curation of seven single-cell RNA-seq studies spanning six tissues: `adipose` [Emont et al., 2022, Hildreth et al., 2021], `bone` [He et al., 2021], `eye` [Gautam et al., 2021], `lung` [Adams et al., 2020], `ovary` [Wagner et al., 2020], and `testis` [Shami et al., 2020]. The benchmark was designed as a carefully linked evaluation set rather than a comprehensive survey of single-cell reporting styles. For each study, a curator read the manuscript and recorded each marker gene–cell type association, its verbatim sentence or figure evidence, source type, linked differential-expression table when available, organism, and Ensembl gene identifier. We preserved the original flat records and converted them losslessly into separate text statements, image statements, and differential-expression tables. Repeated source rows remain traceable through record identifiers rather than being discarded. Ten mouse records from the Adams study remain in the all-organism benchmark totals; the current extraction evaluation is restricted to human records.	`analysis/build_benchmark_evidence.py` `analysis/artifacts/benchmark_evidence_v1/reconciliation.json`
32	`methods-deg-parsing`	Each study's supplementary tables were parsed to extract differential expression gene lists. Header detection was performed automatically by scanning the first 10 rows for a row containing expected column names (e.g., gene, p_value, avg_log2FC). The resulting DEG lists were stored in the same evidence schema as human annotations, enabling direct comparison.	`analysis/benchmark_verification.ipynb`
33	`methods-verification`	Every source row round-tripped exactly and every source-file hash was unchanged. Of 166 primary text evidence claims, 158 (95%) were anchored to exact manuscript offsets; the remaining eight are retained with an explicit unanchored status rather than treated as verified text.	`analysis/artifacts/benchmark_evidence_v1/reconciliation.json` `analysis/build_benchmark_evidence.py`

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#	ID	Claim	Sources
34	`methods-marker-strength-matching`	To quantify where human-curated marker genes fall in the differential expression ranking, we matched each marker to its corresponding entry in the dataset's DEG list. For human-curated markers, matching was performed on the four-part key (dataset, data_id, group_name, feature_id), requiring exact agreement on the cell type, gene, and supplementary data source. For LLM-generated markers, which have no data_id, we matched on (group_name, feature_id) against all DEG data sources within each dataset. Markers that could not be matched—because the gene was absent from the DEG list or lacked an Ensembl ID—were excluded from this analysis.	`analysis/gen_strength.ipynb`
35	`methods-marker-rank-summary`	For each matched marker, we recorded its rank within the cell type's DEG list (rank 1 = highest log-fold change) and its log-fold change. We computed the fraction of markers falling within the top N DEGs for thresholds $N \in \{10, 25, 50, 100\}$ and separately tabulated these fractions for text-sourced markers, image-sourced markers, and generated markers.	`analysis/lfc_comparison.ipynb`
36	`methods-marker-permutation`	To test whether discussed markers are drawn from stronger DEGs than expected by chance, we performed permutation tests at the level of individual cell types within each dataset and data source. For each cell type with at least two discussed markers, we sampled a size-matched set of genes from the top 100 significant ($p < 0.05$) DEGs that were not in the human-curated marker set. We then compared the mean rank of the discussed markers against the null distribution obtained by randomly partitioning the combined set 10,000 times. Tests were two-sided at $\alpha = 0.05$.	`analysis/lfc_comparison.ipynb`
37	`methods-optimal-deg-depth`	To estimate the optimal DEG list depth for recovering each marker set, we computed precision–recall curves by varying a rank threshold N from 1 to 20,000. At each threshold, we defined the top N DEGs as predicted markers and computed precision, recall, and the F-score. The optimal N was defined as the threshold that maximized the F-score. This analysis was performed separately for each dataset and marker source (text, image, or generated).	`analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
38	`methods-marker-source-tests`	Differences in rank and log-fold change between marker sources were assessed using two-sided Mann–Whitney U tests.	`analysis/lfc_comparison.ipynb`
39	`methods-llm-run-settings`	We evaluated three approaches to LLM-based marker gene curation across the seven benchmark datasets. All experiments used Claude (claude-opus-4-6) with temperature 0 and a maximum output length of 32,000 tokens; we did not benchmark model-to-model variation. Costs were computed from the model's API pricing ($5 per million input tokens, $25 per million output tokens for claude-opus-4-6).	`analysis/build_fig2_marker_recovery.py` `analysis/selection_analysis.ipynb`
40	`methods-llm-generation`	In the generation experiment (celltypes-to-genes), we gave the model the list of cell type names from each dataset's human-curated marker set and asked it to generate 5–15 well-established marker genes per cell type. The model received only the species and cell type names—no manuscript text, figures, or supplementary data. In a second generation experiment (genes-to-celltypes), we extracted the gene list for each cell type from the human-curated set, replaced the cell type names with anonymous labels (Group 1, Group 2, . . .), and asked the model to predict the cell type for each group. We measured prediction accuracy as the fraction of exact string matches between predicted and original cell type names. Non-exact predictions were counted as related when they showed substring containment or shared label tokens; remaining predictions were counted as having no lexical relation.	`analysis/ext_vs_gen.ipynb`

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#	ID	Claim	Sources
41	`methods-llm-selection`	In the selection experiment, we parsed each dataset's DEG tables, ranked genes per cell type by absolute log fold change, and presented the top N genes (with log fold change and adjusted p-value) to the model, asking it to select which are true markers. We swept $N \in \{10, 20, 30, 40, 50, 100, 200, 300, 400, 500\}$ for each dataset and reported the N that maximized pair-level F1. As a control for whether cell type identity aids selection, we repeated the full sweep with cell type names replaced by anonymous cluster labels (Cluster 1, Cluster 2, . . .), preserving the gene lists and statistics but removing biological context from the names.	`analysis/selection_analysis.ipynb` `analysis/build_fig2_marker_recovery.py`
42	`methods-llm-extraction`	In the joint extraction experiment, Claude received the full manuscript and a catalog mapping each candidate `data_id` to the cell type labels present in that paper's DEG sources. For every reported human marker association, the model returned the cell type, gene, verbatim supporting text, and one candidate `data_id`, or null when no source was supported. The model did not receive the human annotations. When it returned null, an exact target-label match supplied the narrowest candidate source; this affected five of 152 retained claims. We converted the output to the current `mrkr` claim structure and coalesced records with the same source text and target into marker panels.	`analysis/evaluate_mrkr_joint_deg_extraction.py` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv`
43	`methods-llm-evaluation-subset`	The extraction evaluation used positive human text annotations. Validation required every retained claim to have an exact manuscript offset and every retained marker gene to occur explicitly in that source text. Unsupported genes and claims with ambiguous source offsets were excluded. Gene labels were mapped programmatically with the packaged `gmap.txt`. We computed macro-averaged precision, recall, and F1 across studies for exact normalized cell type–gene pairs and exact `data_id`–cell type–gene triples. The same extracted terms were used for both scores.	`analysis/evaluate_mrkr_joint_deg_extraction.py` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_summary.json` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_pair_review.tsv` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_triple_review.tsv`
44	`methods-llm-source-verification`	The joint experiment saved each raw model response before validation. Every derived artifact records hashes for the prompt, manuscript, human benchmark, DEG catalog, and gene map. The run is therefore resumable and fails if an input changes.	`analysis/evaluate_mrkr_joint_deg_extraction.py` `analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_report.md`
45	`methods-biorxiv-mirror`	We selected bioRxiv preprints from a local mirror of 428,741 MECA archives downloaded from the bioRxiv source repository in January 2025. Each archive contains JATS XML and structured metadata.	`analysis/biorxiv_summary.ipynb` `analysis/build_marker_corpus.py`
46	`methods-biorxiv-filter`	These filters produced 504 bioRxiv manuscripts.	`analysis/biorxiv_summary.ipynb` `analysis/build_marker_corpus.py`
47	`methods-hca-publication-corpus`	Of 515 linked publication records, 475 had manuscript text that could be normalized to Markdown and processed.	`analysis/compile_hca_human_publications.py` `analysis/download_hca_manuscripts.py` `analysis/unify_hca_manuscripts.py` `data/hca/manuscripts_manifest.tsv`

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#	ID	Claim	Sources
48	`methods-corpus-mrkr-workflow`	For each paper, `mrkr extract` returned exact source text, normalized labels, and term offsets. `mrkr ground` mapped human genes with the packaged gene map, organisms to NCBI Taxonomy, cell types and comparisons to Cell Ontology, and tissues to UBERON. `mrkr validate` then checked the source hash, source offsets, explicit term offsets, required term counts, gene direction, and grounding metadata.	`analysis/mine_corpus.py` `analysis/mrkr.lock` `analysis/artifacts/mrkr_corpus_onto_v2.tar.gz.sha256`
49	`methods-corpus-repair`	We repaired these files without another model call. The repair removed only the failing statement or conflicting gene term, recorded every removal, and reran grounding and validation.	`analysis/repair_corpus.py` `README.md`
50	`methods-matched-label-controls`	For each pair of papers that used the same cell type label, we sampled one pair with different labels. Control panels matched the original corpus collection and the base-2 marker-count bin. We required different article keys and repeated the control sampling with 25 fixed random seeds.	`analysis/analyze_mrkr_corpus.py` `analysis/artifacts/mrkr_corpus_analysis_v1/matched_background_seed_sensitivity.tsv` `analysis/analyze_mrkr_corpus.py` `analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_matched_bootstrap.tsv` `analysis/artifacts/mrkr_corpus_analysis_v1/stable_identifier_linkage_summary.tsv`
51	`methods-figure-one-examples`	The matrices in **Supplementary Figure S2** are schematic. The myeloid examples are source-grounded marker statements selected to illustrate relations between reported labels and marker gene panels. No corpus estimate is computed from these examples.	`docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint.tex` `docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint_body.tex`
52	`methods-cap-liver-expression-analysis`	We used CAP dataset 1440 as the local liver myeloid analysis and CAP dataset 1437 as the broader all-cell liver analysis. The files are nested, not independent. All 40,821 cell barcodes in dataset 1440 occur in the 167,598-cell all-cell file.	`analysis/artifacts/cap_liver_local_global_deg_summary.tsv` `analysis/build_cap_liver_local_global_deg.py`
53	`methods-formalization-setup`	We encoded the formal results in Lean 4 under `analysis/formal`. The binary model records studies, partitions, cell groups, pairwise comparisons, marker genes, and marker gene panels. The score model also records real-valued marker scores, thresholds, weighted backgrounds, and sufficient statistics.	`analysis/formal/MarkerIdentifiability/Basic.lean` `analysis/formal/CLAIMS.md` `analysis/formal/lakefile.lean` `analysis/formal/lean-toolchain` `docs/paper/biorxiv/build/supplementary-note.lea`
54	`methods-formalization-theorems`	The Lean development proves the marker-family intersection theorem, monotonicity under restriction of the comparison groups, local-to-global and hierarchy results, equivalence between the all-pair conjunction and the binary marker-family definition, marker panels as hitting sets, panel monotonicity, weighted-background identities, score-thresholded results, and examples in which a cell type–gene projection loses comparison information.	`analysis/formal/MarkerIdentifiability/Basic.lean` `analysis/formal/CLAIMS.md` `docs/paper/biorxiv/supplementary-note.tex`

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